BEE 4750 Homework 2: Systems Modeling and Simulation

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Due Date

Thursday, 09/25/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to draw a systems diagram and identify the type of a feedback.
- Problem 2 asks you to model contaminant concentrations in a river and use simulation to compare the concentrations to a regulatory standard.
- Problem 3 asks you to explore the implications of an ice-albedo feedback in the Earth's climate system by understanding the equilibria of the modeled system and their stabilities.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```julia
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

```
  Activating project at `~/hw2-eliana`
```

```julia
using Plots
using LaTeXStrings
using CSV
using DataFrames
using Roots
```

Problems (Total: 30 Points)

Problem 1 (5 points)

Draw a systems diagram for the relationship between global mean temperature, atmospheric CO2 concentrations, and ocean CO2 concentrations. What are the signs of the interactions between these components and why? What does this suggest about the overall feedback between temperature and the ocean carbon cycle?

Systems Diagram

Feedback Loops Between Atmospheric CO , Temperature, and Oceanic CO

- Atmospheric CO → Temperature (Positive Feedback)

An increase in atmospheric CO concentration leads to an increased greenhouse effect and, therefore, a rise in global mean temperature.
ACO2 (+) → Temp

- Temperature → Atmospheric CO (Negative Feedback)

An increase in global mean temperature will lead to lower concentrations of CO held in the ocean.
- This is due to **Henry's Law**: (k_H) (Henry's constant) decreases with increasing temperature, reducing the solubility of CO in water.

Temp (-) → CO2

- Atmospheric CO → Oceanic CO (Positive Feedback)

An increase in atmospheric CO increases oceanic CO due to higher partial pressure and an increased flux from atmosphere to ocean.
CO2 (+) → OCO2

- Oceanic CO → Atmospheric CO (Negative Feedback)

An increase in oceanic CO removes CO from the atmosphere, lowering atmospheric levels.
- Conversely, a decrease in oceanic CO (e.g., due to increased temperature) releases CO into the atmosphere, raising atmospheric levels.

OCO2 (-) → ACO2

Overall System Behavior

The system is primarily a **positive feedback loop**:
- Increases in atmospheric CO → higher global mean temperature → decreases in oceanic CO → increases in atmospheric CO .

There is also a **dampening effect**:
- An increase in oceanic CO uptake reduces atmospheric CO levels, partially offsetting the positive feedback.

> **Tip**
>
> Think about Henry's law for CO2.

Problem 2 (15 points)

A river which flows at 10 km/d is receiving discharges of wastewater contaminated with CRUD from two sources which are 15 km apart, as shown in the Figure below. CRUD decays exponentially in the river at a rate of 0.36 d^{-1} and is deposited by the atmosphere along the river at a rate of 54 kg/km/d.

Schematic of the river system in Problem 2

Problem 2.1

Draw a systems diagram with the relevant control volume(s) denoted and any relevant in/outflows between these boxes. How did you decide how many boxes were needed?

Problem 2.2

Develop a model for the concentration of CRUD downriver by formulating and solving the appropriate differential equation(s) analytically.

> **Tip**
>
> Formulate your model in terms of distance downriver, rather than leaving it in terms of time from discharge.

Problem 2.3

Determine if the system in compliance with a regulatory limit of 2.3 kg/(1000m^3). You can do this analytically or computationally.

Problem 3 (10 points)

In class, we discussed the ice-albedo feedback and its possible influence on melting the hypothesized "Snowball Earth". In this problem, we'll introduce a simple model of the Earth's energy balance with includes this feedback and examine the stability of the climate.

This simple model of the energy balance (averaged over the entire planet) is:

$$\underbrace{C\frac{dT}{dt}}_{\text{change in heat}} = \underbrace{\frac{(1-\alpha)S}{4}}_{\text{incoming radiation}} - \underbrace{(A-BT)}_{\text{outgoing radiation}} + \underbrace{a\ln\left(\frac{[CO_2]}{[CO_2]_{PI}}\right)}_{\text{greenhouse effect}}, \tag{1}$$

where:

- T is the Earth's global mean temperature (in °C);
- C is the heat capacity of the atmosphere and shallow ocean, taken to be 51 J/m^2/°C;
- α is the planetary albedo, or the fraction of incoming radiation reflected by the Earth, which has a present-day value of approximately 0.3;
- S is the solar constant, or the amount of solar radiation received by the Earth averaged over area, which has a present-day value of 1368 W/m^2 but during the Neoproterozoic Era had a value of 1272 W/m^2 (this is divided by four because the Earth is a sphere but S is the radiation captured by a disc with radius equal to that of the Earth);

- A and B are coefficients from the linearization of outgoing radiation physics, and have corresponding values $B = -1.3$ W/m^2/°C (estimated from a number of lines of evidence about the sensitivity of outgoing radiation to temperature; this is negative due to a sign convention about the direction of incoming vs. outgoing radiation) and $A = 221.2$ W/m^2 (estimated by assuming the pre-industrial temperature of 14°C was stable without anthropogenic greenhouse gas emissions).

We will ignore the greenhouse effect term as we are considering the Earth system well before humans were around.

Problem 3.1

Discretize the climate model (Equation 1) using forward Euler integration and a time step of $\Delta t = 1$ yr.

T(t + Δ t) = T(t) + $\frac{\Delta t}{C}$ ((1- α) * ($\frac{S}{4}$) - A - B*T(t))

Problem 3.2

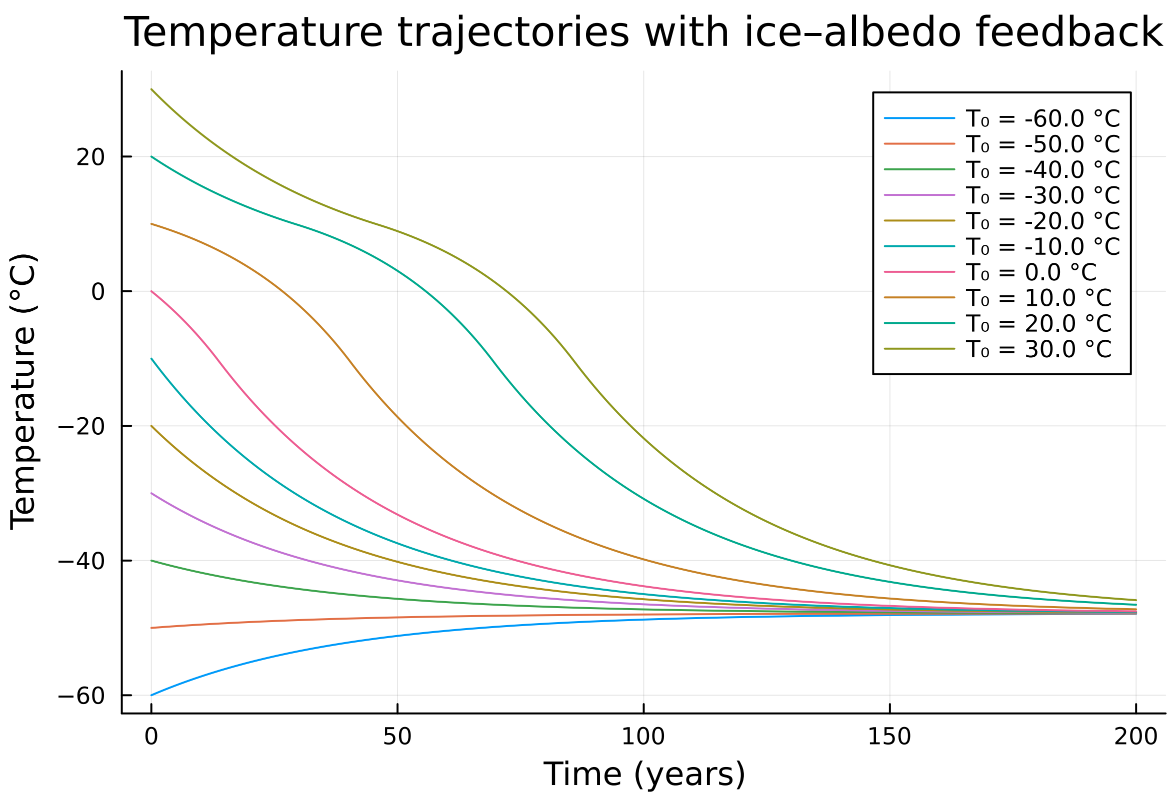
Rather than assuming a constant value for the albedo α, we will represent the ice-albedo feedback by letting α depend on T:

$$\alpha(T) = \begin{cases} \alpha_i & \text{if } T \leq -10°\text{C} \\ \alpha_i + (\alpha_0 - \alpha_i)((T + 10)/20) & \text{if } -10°\text{C} \leq T \leq 10°\text{C} \\ \alpha_0 & \text{if } T \geq 10°\text{C}. \end{cases}$$

Let $\alpha_i = 0.5$ and $\alpha_0 = 0.3$. Simulate the simple climate model using the Neoprotereozoic value of S with temperature-varying albedo for initial values of T spanning $-60°\text{C} \leq T_0 \leq 30°\text{C}$ over a period of 200 years. Plot the temperature trajectories. How many equilibria are there and what are their stabilities?

```
using Plots

#Constants
#------------------------------
C = 51 # (J/m^2/°C) the heat capacity of the atmosphere and shallow ocean
alpha_i = 0.5 #Initial planetary albedo
alpha_0 = 0.3
S_n = 1272 #(W/m^2) Solar constant for Neoproteorozoic Era
S_p = 1368 #(W/m^2) Solar constant for Present day
B = 1.3 #(W/m^2/°C)
A = 221.2 #(W/m^2)
```

```julia
#Function for albedo
#------------------------------
function albedo(T)
    if T <= -10
        alpha = alpha_i
    elseif T >= 10
        alpha = alpha_0
    else
        alpha = alpha_i + (alpha_0 - alpha_i)*((T + 10)/20)
    end
    return alpha
end

#Finding temperatures
#------------------------------
initial_temps = range(-60, stop=30, length=10)
years = 0:200
dt = 1
temps_changes = zeros(length(years), length(initial_temps))
for (i, T0) in enumerate(initial_temps)
    T = T0
    temps_changes[1,i] = T
    alpha = albedo(initial_temps[i])
    for j in 2:length(years)
        alpha = albedo(T)
        T = T + (dt/C) * ((1 - alpha) * (S_n / 4) - A - B * T)
        temps_changes[j,i] = T
    end
end

#PLot!
#---------------------------------------
plot(years, temps_changes[:, 1], label="T  = $(round(initial_temps[1], digits=1)) °C")
for j in 2:length(initial_temps)
    plot!(years, temps_changes[:, j], label="T  = $(round(initial_temps[j], digits=1)) °C")
end

xlabel!("Time (years)")
ylabel!("Temperature (°C)")
title!("Temperature trajectories with ice-albedo feedback")
```

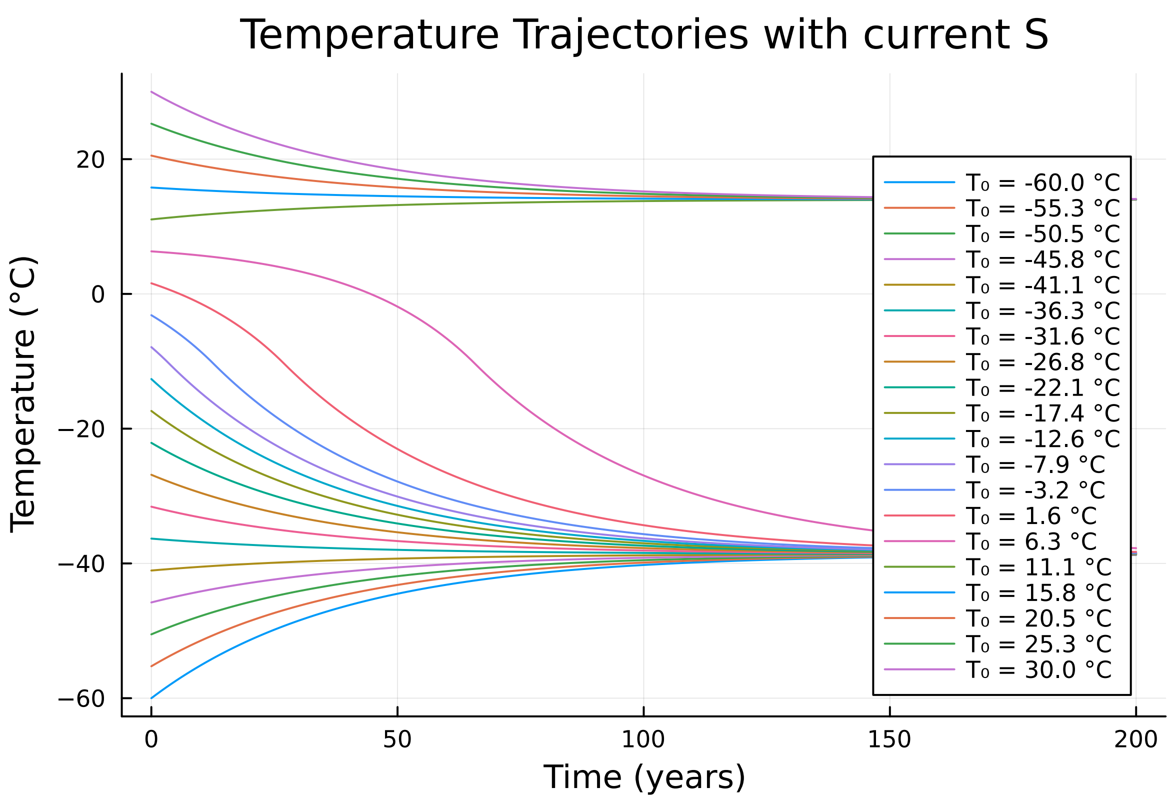


Problem 3.3

One might hypothesize that one cause for the transition from Snowball Earth (the stable frozen equilibrium from Problem 3.2) was an increasing amount of incoming solar radiation (as expressed by an increase in S). Examine this hypothesis using our simple model. Is this factor enough to cause the planet to warm to the pre-industrial temperature of 14°C?

```
#Finding temperatures for current Solar...
#------------------------------
initial_temps = range(-60, stop=30, length=20)
years = 0:200
dt = 1
temps_changes = zeros(length(years), length(initial_temps))
for (i, T0) in enumerate(initial_temps)
    T = T0
    temps_changes[1,i] = T
    alpha = albedo(initial_temps[i])
    for j in 2:length(years)
        alpha = albedo(T)
```

```
        T = T + (dt/C) * ((1 - alpha) * (S_p / 4) - A - B * T)
        temps_changes[j,i] = T
    end
end

#PLot!
#---------------------------------------
plot(years, temps_changes[:, 1], label="T  = $(round(initial_temps[1], digits=1)) °C")
for j in 2:length(initial_temps)
    plot!(years, temps_changes[:, j], label="T  = $(round(initial_temps[j], digits=1)) °C")
end

xlabel!("Time (years)")
ylabel!("Temperature (°C)")
title!("Temperature Trajectories with current S")
```



Just an increase in the solar radiation isn't enough to bring the temperature up to the 14°C we saw in pre industrial times. We can see in the plot above that there is still the lower equillibrium around -40°C and if the temperature was anywhere between $10 and - 6^{\circ}$C it would have just gone back to that lower equillibrium. There would've had to of been a dramatic increase in temperature to above 10°C and then it would have gone to the upper equilibrium of 14°C.

References

I used ChatGPT to help with latex formatting of text in markdown